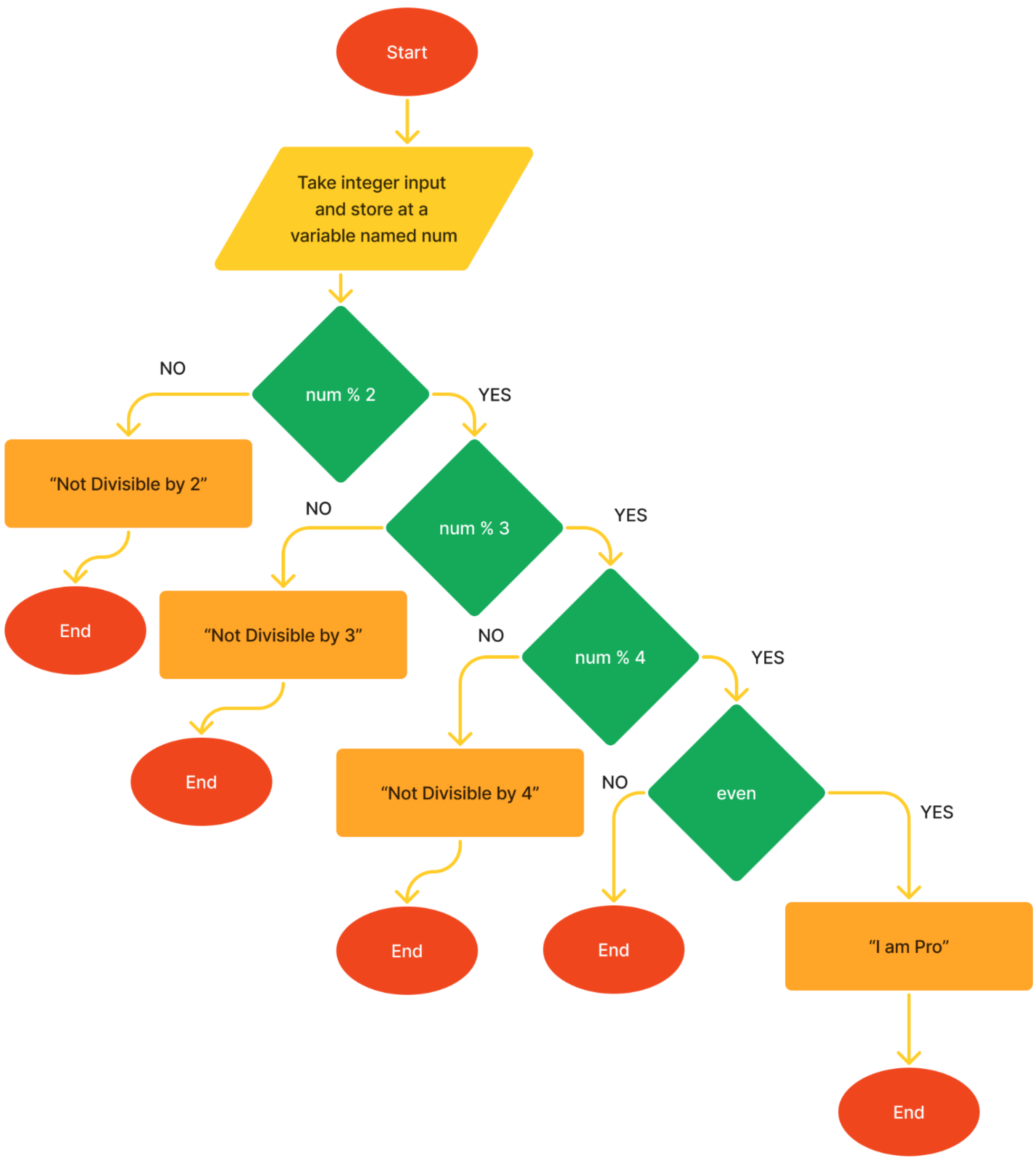
Week-2 (Practice Problems)
Branching

(1)
Convert the given **flowchart** to python code.



(2)
Users will always enter a **4 digit number** and you need to add the first digit and the last digit of that number and see if the summation is even or not. If the summation is even then print **“Even”** otherwise print **“Odd”**.

[YOU ARE NOT ALLOWED TO USE STRINGS]

Input	Output
1756 **Explanation:** **First digit is 1 and Last digit is 6. Now if we add 1+6 we get 7, and 7 is Odd**	**Odd**
2424 **Explanation:** **First digit is 2 and Last digit is 4. Now if we add 2+4 we get 6, and 6 is Even**	**Even**

(3)
You want to join a Cycling community, to join there you need to enter the distance traveled by cycle in **km**, using that distance you will either get a title or you will be rejected to join the club. The table below shows the title for each distance in **m**.

[HINT: Read the question carefully]

Distance (m)	Result
>10000	“Veteran Cyclist”
>=9000	“Hardcore Cyclist”
>=8000	“Pro Cyclist”
>=7000	“Amateur Cyclist”
>=6000	“Beginner Cyclist”
>=5000	“Rookie Cyclist”
<5000	“You are not Eligible”

Input	Output
12	**Veteran Cyclist**
4.5	**You are not Eligible**

(4)
You already have two angles **a** and **b** with values **85** and **35** respectively in two different variables. Now, the user will enter the **third angle**, after that you have to say whether you can form a triangle with these three angles.

Input	Output
60	**Triangle can be formed**
30	**Triangle can not be formed**

(5)

Write a program that asks the user for an hour as input and prints a message, which lab class is taking place in the lab based on the following table. If the user enters any value that is not in the table, your program should print **“Wrong Input”**.

Hour (input)	**Message (output)**
8	**1st lab**
9	**1st lab**
10	**1st lab**
11	**2nd lab**
12	**2nd lab**
13	**2nd lab**
14	**3rd lab**
15	**3rd lab**
16	**3rd lab**

Input	**Output**
13	**2nd lab**
16	**3rd lab**
7	**Wrong input**

(6)

Suppose the following expressions are used to calculate the values of **P** for different values of **Q**:

$$P = 6000 + 15Q^2 \quad if\ Q < 50$$

$$P = \frac{1000}{3+(Q^2/1900)} \quad if\ Q >= 50$$

Write a python code that takes the value of **Q** as an user input and shows the value of **P** as an output.

(7)
Write a python program that asks the user for hours in **integers** and prints the **CEFR Level** according to the table below. If the user enters a number of hours not included in the table below, print **"Unknown"**.

CEFR Level	**Number of Hours (Inclusive)**
C2	**1,000—1,200**
C1	**700—800**
B2	**500—600**
B1	**350—400**
A2	**180—200**

Input	**Output**
1150	**C2**
650	**Unknown**